Project Report

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Branch: MCA **Section/Group: 3(A)**
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Subject Name: Internet of Things Lab **Subject Code: 24CAP-751**

Aim: The aim of this project is to develop a smart system that can detect gas leakage. It also automatically controls electrical devices to ensure safety during unsafe conditions.

Objectives:

- To design a gas detection system using IoT technology
- To monitor gas levels in real-time using an MQ-2 sensor
- To implement automatic safety actions when gas exceeds a threshold
- To control devices using a relay module
- To provide visual indication using an LED
- To enhance safety in homes and industries

Components Required:

Sno	Name of Component	Qty.
1.	NodeMCU (ESP8266)	1
2.	Gas Sensor (MQ-2)	1
3.	Relay	1
4.	LED	3
5.	Resistor (230 Ohm)	3
6.	Breadboard	1

Details of Components:

1. NodeMCU:

The NodeMCU board is a low-cost, open-source IoT platform widely recognized for its user-friendly approach to building Internet-connected projects. Based on the ESP8266 WiFi chip, it combines a microcontroller with WiFi capabilities, enabling developers to create connected devices with ease.

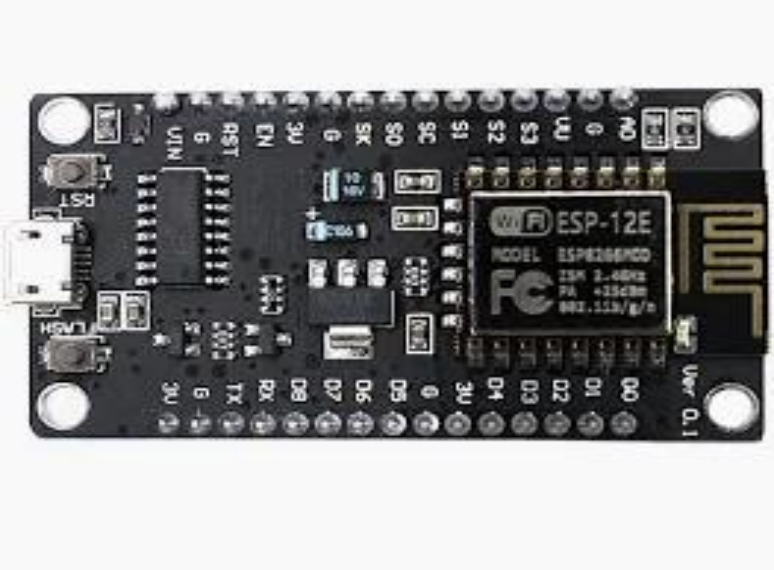

Figure 1 : NodeMCU Board

2. Gas Sensor:

The gas sensor is used to detect the presence and concentration of gases in the environment. It helps identify harmful or combustible gases such as LPG, methane, smoke, and carbon monoxide. In safety systems, a gas sensor continuously monitors air quality and sends signals when gas levels exceed a safe limit, allowing the system to take automatic actions like triggering alerts or turning off electrical devices.

Figure 2: MQ2 Sensor

3. Relay:

A relay is an electrically operated switch used to control high-power devices using a low-power signal from a microcontroller. It allows safe switching of appliances by turning them ON or OFF automatically based on the system's output, without direct human intervention.

Figure 3: Relay

Block Diagram of Designed Model:

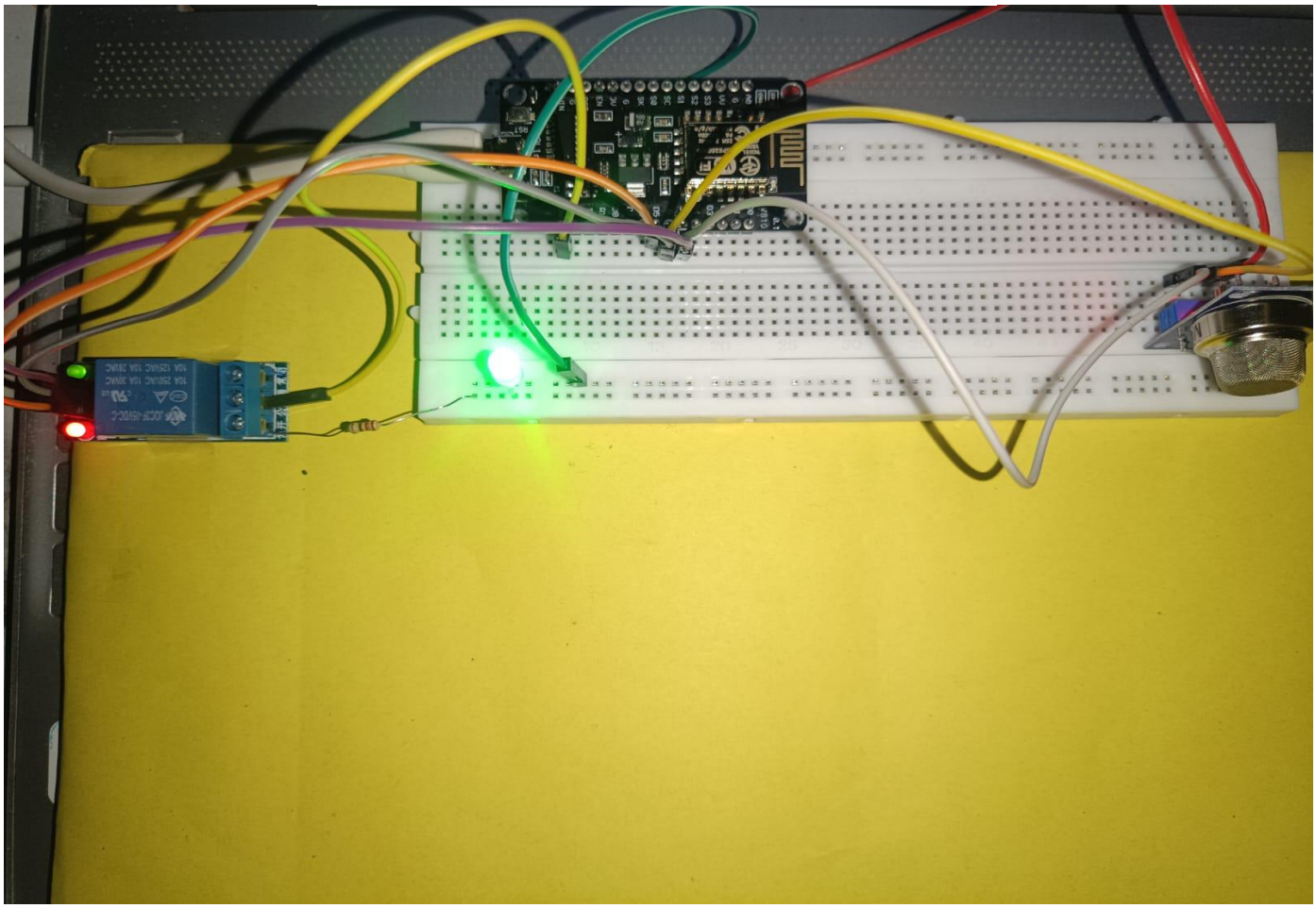

Figure 4: Designed Model

Working of Designed Model:

The system works based on gas detection and threshold comparison.

1. The MQ-2 sensor continuously detects gas levels in the environment.
2. The sensor sends analog data to the NodeMCU.
3. The NodeMCU compares the sensor value with a predefined threshold value set in the code.
4. If the gas level exceeds the threshold:
    - The LED turns OFF (or indicates alert depending on logic)
    - The relay turns OFF, cutting power to connected devices
5. If the gas level returns to normal:
    - The LED turns ON
    - The relay turns ON, restoring normal operation

This automatic switching ensures safety without manual intervention.

Output of Deigned Model/Prototype

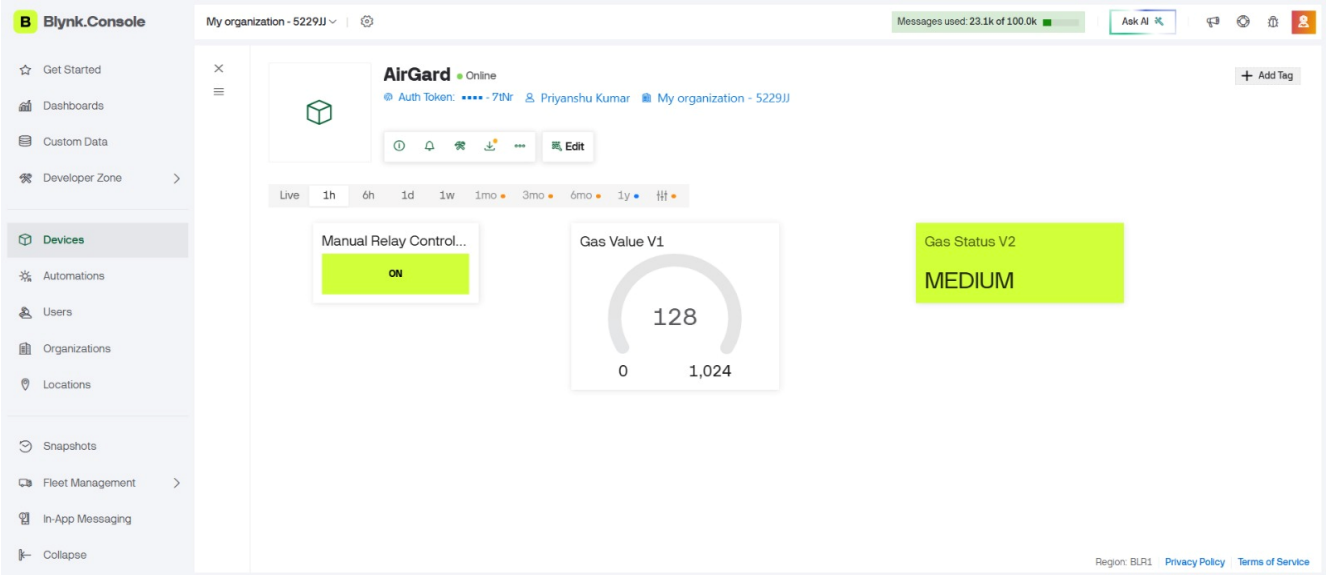

Figure 5: Blynk Output

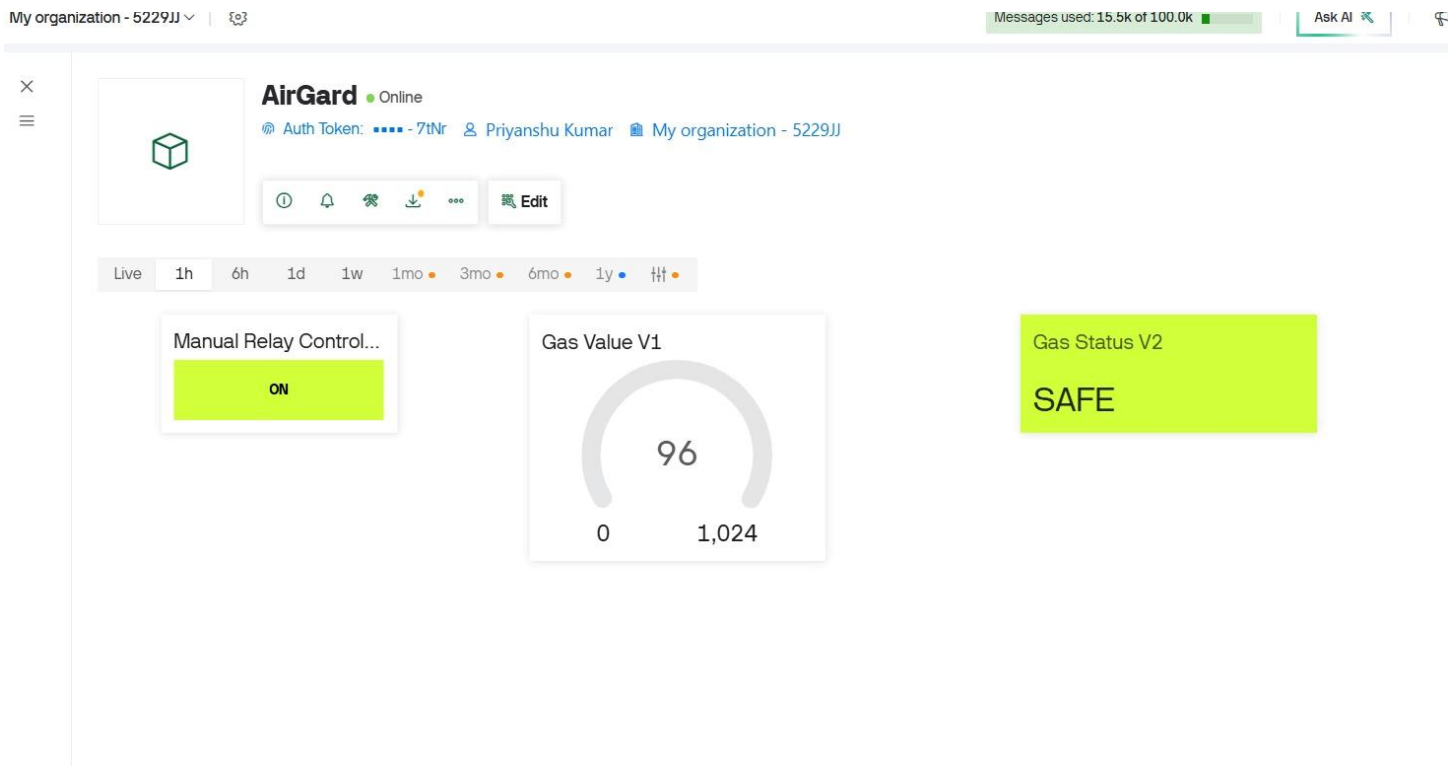

Figure 6 : Output on BLYNK Cloud

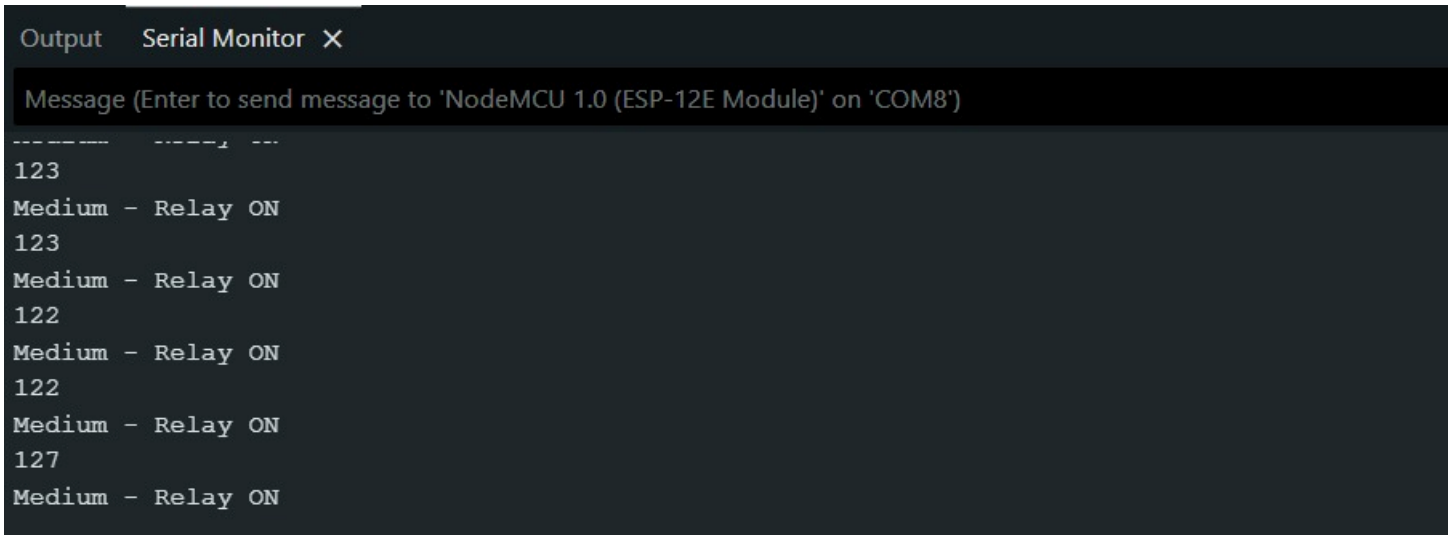

Figure 7 : Serial Monitor Output

Learning outcomes (What I have learnt):

- Understanding of IoT-based safety systems using sensors and microcontrollers
- Knowledge of working with NodeMCU and MQ-2 gas sensor
- Ability to design and implement automatic control systems using a relay
- Hands-on experience with circuit connections and hardware components
- Improved skills in programming using Arduino IDE for real-time applications